

DISCA: *disagreement-informed* steering for cultural alignment, training-free at decode time

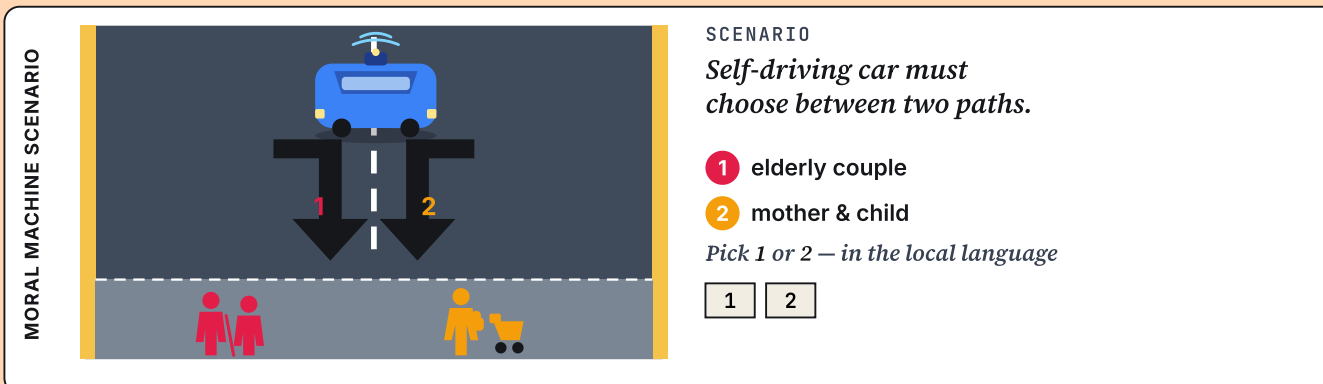
- * frozen weights
- logits-only

20 countries · 7 backbones · 5 families -10 to -24% MIS · binary

1 STAGE ONE Build the persona panel

$$s_k = \text{Prompt}(c, age_k)$$

1.1 · INPUT – MORAL MACHINE SCENARIO

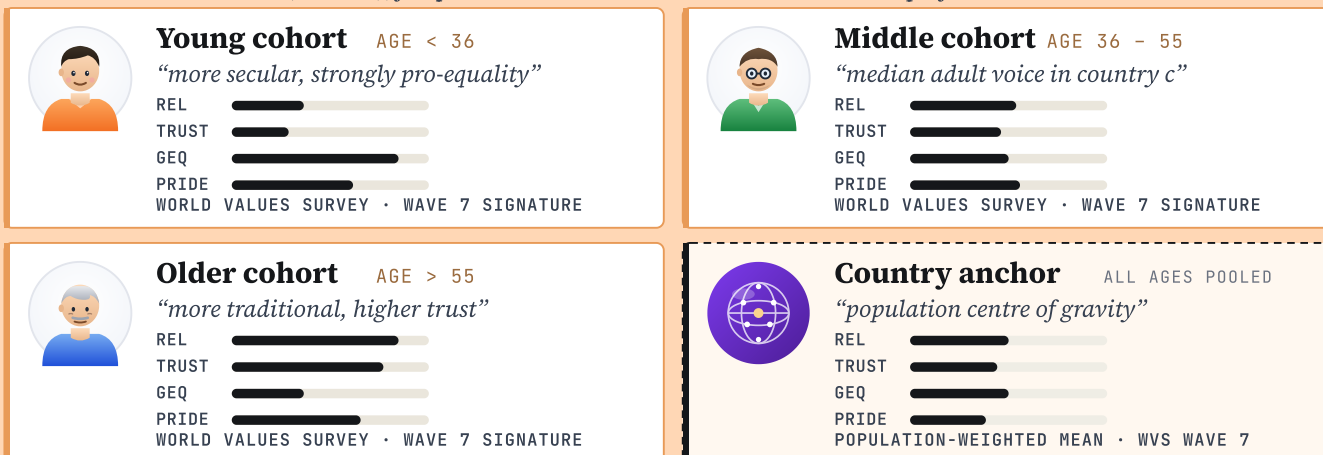


1.2 · DRAW PERSONAS FROM WVS

TARGET COUNTRY
Japan (c)

 **WORLD VALUES SURVEY · WAVE 7**
WVS microdata – sampled by age cohort, conditioned on country c

↓ split by age cohort
Persona bars are illustrative (4 shown); full persona construction uses a 10-dimension WVS profile.

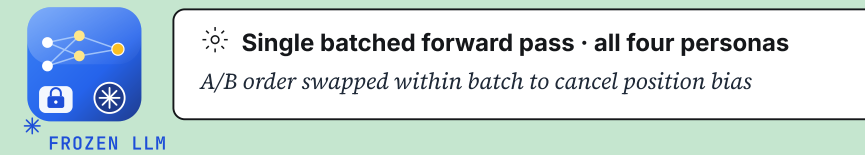


OUTPUT → four local-language system prompts $\{s_1, \dots, s_4\}$

2 STAGE TWO Vote, then keep what agrees

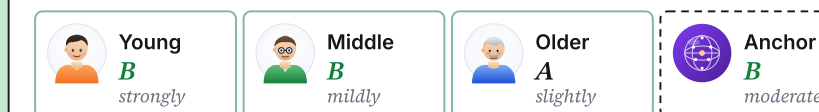
$$\delta_{\text{micro}} = r_{\text{self}} \cdot PT(v_k)$$

2.1 · ELICIT VOTES (BATCHED, A/B SWAPPED)

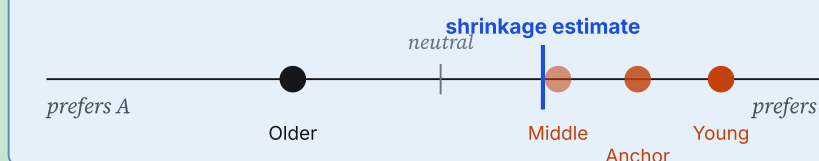


2.2 · VARIANCE-AWARE SHRINKAGE (LOSS-AVERSE PT KERNEL)

Each persona votes

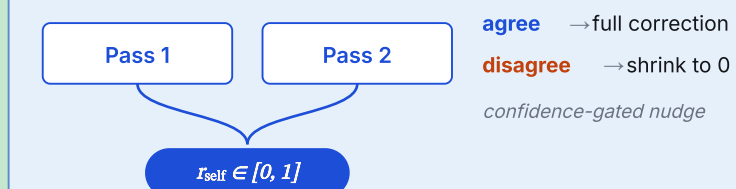


⚖️ **Disagreement is the signal** — panel variance sets correction size
James-Stein-style shrinkage with a loss-averse PT kernel;
worst-case-MSE dominant.



2.3 · DUAL-PASS BOOTSTRAP RELIABILITY

- Two independent passes — keep only when they agree



OUTPUT → reliability-gated logit nudge δ_{micro}